

On the Design of Mobility-Aware Systems: A Tourist's Perspective

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Abstract—The design of effective mobility-aware systems requires a deep understanding of how different user groups navigate and interact within urban environments, enabling the development of services and infrastructures that adapt to dynamic movement patterns. Tourist mobility patterns, in particular, offer valuable insights into the spatial and social dynamics that influence system design decisions for smart cities, recommendation engines, and urban planning applications. This work presents a comprehensive approach to modeling urban mobility networks by examining tourist behavior through network analysis techniques. Using Rome as a case study, we leverage a large dataset of Foursquare check-ins collected over 18 months to construct a co-visitation graph where nodes represent points of interest and edges reflect shared visits by the same users. Through network analysis of Rome's tourist mobility, our work identifies clear structural differences between visitor types: long-term visitors form cohesive, highly clustered mobility networks, while short-term visitors create more fragmented interaction patterns. Key points of interest (especially transportation hubs and cultural landmarks) serve as essential connectors shaping network flow, following preferential attachment dynamics. Mobility-aware systems play a crucial role in leveraging such insights to design adaptive, data-driven services that respond to varying mobility behaviors, ultimately enhancing urban efficiency and user experience.

Index Terms—mobility-aware system, network science, tourism, location-based social network, mobility.

I. INTRODUCTION

Tourist mobility in urban environments presents unique challenges and opportunities for understanding spatial and social dynamics. With the growing availability of data from Location-Based Social Networks (LBSNs) [1], it has become possible to analyze movement patterns, identify mobility hubs, and uncover latent structures that govern the flow of people within cities. Thus, managing urban mobility goes beyond traditional transportation systems and includes the study of how users interact with urban spaces through visits to points of interest (PoIs), among other aspects.

This work investigates the mobility dynamics in the touristic city of Rome by analyzing a real-world dataset of Foursquare check-ins over an 18-month period [2]. By modeling a complex network where nodes represent PoIs and edges encode co-visitation patterns by the same users, we explore how tourist activity influences urban interaction networks. The presence of high-centrality nodes, such as transportation hubs and cultural

landmarks, plays a central role in defining short path lengths, high clustering, and a scale-free topology [3], which are key indicators of a mobility-critical infrastructure.

Our main contribution lies in revealing how the structure and semantics of mobility patterns differ depending on user profiles, particularly their length of stay. Through network metrics such as degree distribution, clustering coefficient, betweenness centrality, and diameter, we show that tourists with longer stays contribute to more cohesive and clustered mobility networks, while short-term visitors exhibit fragmented and sparse interaction structures. These insights are particularly relevant for urban mobility planning, infrastructure optimization, and real-time location-aware services in smart cities.

By framing touristic movement as a complex social mobility network, our work offers novel implications for the design of mobility-aware systems and services. Furthermore, it highlights the role of preferential attachment mechanisms in reinforcing the attractiveness of central PoIs, thereby impacting the spatial distribution of crowds and congestion. These findings contribute to the broader discussion on mobility management by showcasing how passive digital traces can be leveraged to inform infrastructure planning and adaptive network strategies in highly dynamic urban environments.

For tourists, network metrics help optimize itinerary planning. High-centrality places serve as anchors for routes, offering engaging paths within limited timeframes. The ability to identify less congested alternatives through structural analysis also enables crowd-aware recommendations, improving the tourist experience. Moreover, shared behavioral patterns across users can be leveraged for social mobility matching, allowing travelers with similar interests to be connected for co-visits or guided tours.

Urban planners benefit from insights into city hotspots and flow bottlenecks. By examining how long-stay and short-stay users interact with PoIs, planners can better allocate different types of resources (e.g., public washrooms and rest areas), transit links, or event zones. Regions of high clustering can be prioritized for tourism investment, while fragmented or underutilized areas may be targeted for revitalization.

For smart city system developers, the degree distribution and centrality metrics inform adaptive transportation logistics (e.g., dynamic routing of buses and shared mobility fleets),

especially during events or peak seasons. Real-time systems can use this structure to feed context-aware alerts, digital signs, or augmented reality tools. Even more, integrating this data into digital twin models allows simulation of emergent crowd behaviors, optimizing urban design under different stress conditions.

In short, this work provides a versatile framework with direct applications in sustainable tourism development, infrastructure optimization, and smart mobility ecosystems (see Fig. 1). By leveraging network science, this work analyzes social dynamics in a typical tourist city through a check-in dataset. More precisely, this work aims to understand how social interactions are affected by touristic attractions. Obtained results suggest that network properties vary according to the type of user activity. In addition, the findings show that specific places tend to attract more people, thereby playing a vital role in the structure of the network.

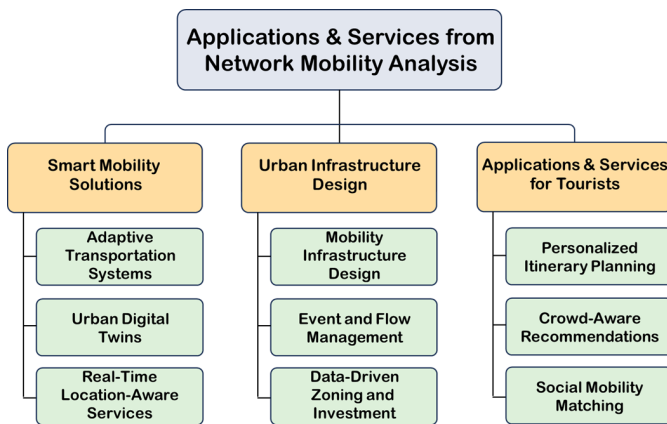


Fig. 1. Analysis of tourist mobility via network science

The rest of this work is organized as follows. Section II presents a review of related work, focusing on key studies involving network analysis in the context of tourism. Section III provides a detailed description of the dataset used. Section IV outlines the network modeling approach, while the methodology adopted to achieve the results is detailed in Section V. Next, the results and their discussion are presented in Section VI. Finally, Section VII offers concluding remarks and suggests possible directions for future work.

II. RELATED WORK

The study of urban mobility and human behavioral patterns has significantly advanced in recent years thanks to the widespread use of LBSNs. These platforms provide rich, fine-grained data about user movement and place visitation, offering unprecedented opportunities to model human mobility patterns, forecast spatial occupancy, and investigate the dynamics of urban space utilization.

Recent studies have applied network analysis to LBSNs in a large number of scientific fields, such as recommendation systems [4], urban planning [5], and social behavior modeling [6]. Despite these significant contributions, the mentioned

studies do not examine how tourist attractions influence the broader dynamics of the city

In the tourism domain, Sun et al. [7] proposed a social media-based methodology to construct and analyze co-visitation networks in peri-urban tourism areas, revealing patterns that can guide planning and visitor experience design. Such networks enable the understanding of how different points of interests are interconnected through human mobility patterns. However, their focus on an emerging urban hub contrasts with our study, which examines a touristic urban environment characterized by a greater diversity of attractions and a strong international tourism component.

Other works have investigated the spatial organization of tourism in large regional contexts. For example, Weng and Li [8] explored ArcGIS and geographic detector models to analyze the spatial distribution and influencing factors of tourist attractions, revealing a concentrated distribution pattern driven primarily by major urban centers and shaped by socioeconomic development, transportation, and natural features. Another study examined the spatial network structure of the tourism economy in the urban agglomeration of the middle reaches of the Yangtze River using a tourism economic gravity model and social network analysis [9]. These studies rely on aggregated official statistics and static geographic data, which limit insights due to low temporal resolution. In contrast, our research uses fine-grained check-in data that captures visitor behaviors and co-visitation patterns at a micro level, enabling deeper analysis of urban tourism dynamics.

More recent approaches have leveraged network-based approaches to better understand tourism dynamics across different contexts. One comparative analysis examined long-term tourist mobility patterns in Indonesia, Thailand, and Vietnam using 20 years of online review data, combining network analysis with association rule mining to uncover spatial concentration trends and culturally driven preferences [10]. Another study focused on coastal tourism, using perception network analysis to map the interrelations among sensory, functional, social, cultural, natural, hospitality, and auxiliary components of the “destination experiencescape” [11]. While these studies provide an in-depth analysis of tourism patterns and experiential components, they do not examine the structural interactions between multiple points of interest – a gap that our study addresses by focusing on co-visitation network structures in the historical center of Rome.

These works provide the methodological and conceptual foundation for the present study, which aims to investigate how tourist attractions shape the structural topology of urban co-visitation networks. Leveraging LBSN data, our research analyzes individuals’ spatiotemporal location patterns and models interactions between PoIs through co-visitation networks to uncover underlying structural relationships using network science metrics.

III. DATASET

To enhance the realism of this study, we employed a large-scale LBSN dataset obtained from Foursquare [2]. This dataset

captures rich spatiotemporal traces of human activity through user-generated check-ins, offering a valuable resource for analyzing urban mobility and visitation patterns. The original dataset contains check-in records from 415 cities across 77 countries, totaling more than 33 million check-ins collected over an 18-month period¹. Each check-in event is associated with detailed attributes that enable both spatial and temporal analysis:

- **User ID:** An anonymized unique identifier for each user;
- **Venue ID:** A unique identifier for each PoI on Foursquare;
- **Timestamp:** The exact date and time of the check-in;
- **Coordinates:** The geographical latitude and longitude of the PoI;
- **Category:** The semantic classification of the PoI, such as restaurant, church, square,

In this study, we focus on a reduced scenario that includes only check-ins made in the central region of Rome, Italy. This area covers the city's main districts, encompassing approximately $9.693 \text{ km} \times 9.052 \text{ km}$, resulting in a total area of 87.74 km^2 . Each record specifies the date and the exact PoI where a user performed a check-in. After filtering, the dataset reveals an average activity level of 54 check-ins per day, with typical fluctuations around 40 daily check-ins (see Fig. 2). Notably, during certain periods in 2012, daily check-ins exceeded 200, suggesting peaks likely associated with tourism seasonality or major events.

The final dataset used in this study comprises 28,654 check-ins performed by 2,849 distinct users, covering 5,697 unique PoIs across the study area.

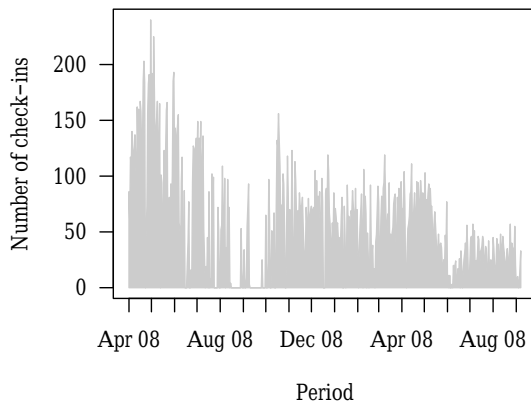


Fig. 2. Total number of check-ins made in the city of Rome.

IV. NETWORK MODELING

Graph theory is widely employed to represent and analyze complex systems across various domains, such as neural networks, social networks, the Internet, among others [12]. This approach provides an intuitive visual representation and facilitates the development of mathematical models that capture the relationships between system elements.

¹From April 2012 to September 2013

In this work, the check-in network was modeled as an undirected graph $G(\mathbb{V}, \mathbb{E})$, where $\mathbb{V} = \{v_1, v_2, \dots, v_n\}$ denotes the set of PoIs, and $\mathbb{E} = \{e_1, e_2, \dots, e_m\}$ represents the set of edges capturing co-visitation relationships between these locations. More precisely, an edge $e_{ij} \in \mathbb{E}$ exists whenever two locations i and j were visited by the same individual. This formulation allows us to represent patterns of movement as a network structure, enabling the application of graph-theoretic metrics to uncover central attractions, connectivity patterns, and clusters of frequently co-visited locations.

For example, an edge connects the nodes representing the *Colosseum* and the *Pantheon* if a tourist visited both locations (see Fig. 3). It is important to note that temporal information from the check-ins was not considered during the graph construction. Furthermore, since a pair of points of interest can share multiple visitors, the graph is modeled as a weighted graph, where edge weights represent the number of shared visitors between the connected nodes.

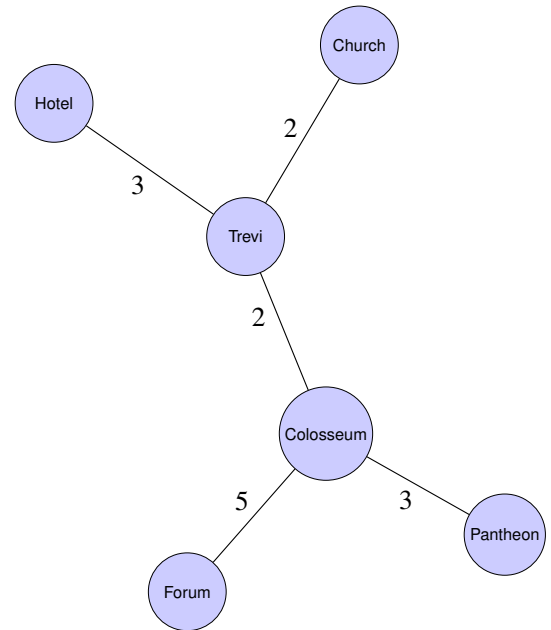


Fig. 3. Example of a co-visitation network.

V. METHODOLOGY

In a touristic city, certain points of interest tend to receive a higher number of visits, resulting in increased attraction to those regions. In this study, we investigate whether the touristic potential of a point of interest influences the popularity of other nearby or related locations. Therefore, this work investigates the following research question:

How do tourist attractions influence the structure and dynamics of co-visitation networks in urban mobility patterns?

To address this question, we formulate the following hypotheses:

- **Null Hypothesis (H_0):** Tourist attractions do not significantly differ from non-tourist locations in terms of

their structural role within the co-visitation network (e.g., degree centrality or influence on network connectivity).

- **Alternative Hypothesis (H_1):** Tourist attractions have a significantly greater structural influence in the co-visitation network than non-tourist locations, exhibiting higher centrality and contributing more strongly to network connectivity.

In this study, we considered the following network metrics:

- **Vertex Degree:** Defined as the number of edges adjacent to a vertex v_i .
- **Average Degree:** The average degree over all vertices in the network, defined as:

$$\bar{k} = \frac{2m}{n} \quad (1)$$

- **Edge Density:** The ratio between the number of edges in the network and the total possible number of edges:

$$\rho = \frac{2m}{n(n-1)} \quad (2)$$

- **Clustering Coefficient:** The ratio between the number of existing connections among the neighbors of a node v_i and the maximum possible number of such connections. The global clustering coefficient is defined as the average of all individual node clustering coefficients.
- **Betweenness Centrality:** Measures how often a node appears on the shortest paths between other nodes. The betweenness centrality of a vertex v_u is given by Equation 3:

$$B_u = \sum_{v_i, v_j \in V} \frac{\sigma_{(i,j,u)}}{\sigma_{(i,j)}} \quad (3)$$

where $\sigma_{(i,j,u)}$ is the number of shortest paths between vertices v_i and v_j that pass through v_u , and $\sigma_{(i,j)}$ is the total number of shortest paths between v_i and v_j .

- **Diameter:** The largest distance between any pair of vertices in the network.
- **Giant Connected Component (GCC) Size:** The number of nodes in the largest connected subgraph of the network. It represents the main cluster where nodes are mutually reachable, serving as a key indicator of the network's overall connectivity and robustness.
- **Average Shortest Path Length:** The average number of steps along the shortest paths for all possible pairs of nodes in the network. It quantifies the typical distance between nodes and reflects the efficiency of information or flow traversal within the network.

For the experiments, a machine running Linux Mint 22, equipped with a quad-core Intel Core i5-7200U processor (2.50 GHz) and 8 GB of RAM, was used. The R programming language [13] was employed for network analysis. The *igraph* library² enabled the construction of the graph and the computation of the evaluated network metrics.

²Available at: <http://igraph.org/r/>

VI. RESULTS AND DISCUSSION

This section presents the results obtained from the analysis of the check-in network of the city of Rome. The objective of the experiments was to address the research question posed in the previous section. Therefore, different complex network metrics were analyzed as a means to understand the structural characteristics of the network and how they respond to different patterns of social activity.

Table I presents the categories that received the most check-ins during the evaluated period. The total number of check-ins indicates a preference for public spaces. The squares of Rome are notably well-known due to their history and architecture, such as the *Piazza di Spagna*. Therefore, they attract visitors from all over the world. The most popular location in the city was the *Colosseum*, which received 94 check-ins. The presence of train stations is also noteworthy. The *Stazione Termini* is the largest train station in Italy and one of the largest in Europe. Approximately 500 thousand people pass through it daily [14]. Although the total number of check-ins provides some interesting insights about people's activity, there are other latent pieces of information that can be analyzed through the network structure.

TABLE I
CATEGORIES OF THE MOST VISITED PLACES.

Ranking	Category	Total check-ins
1	Square	2.866
2	Train Station	1.993
3	Historic Center	1.980
4	Restaurant	1.433
5	Hotel	1.126
6	Coffee Shop	1.076
7	Monument	983
8	Church	951
9	Office	913
10	Residence	688

Fig. 4 depicts the network modeled in Section IV. The constructed network comprises a total of 5,697 nodes and 2,357,497 edges. To facilitate visualization, the figure displays only the minimum spanning tree of the network. The node highlighted in red corresponds to *Stazione Termini*, which exhibits the highest **betweenness centrality** measure (0.2818). This node plays a crucial role by connecting locations across different regions of the network, largely due to its function as a major transportation hub. Fig. 5 summarizes the centrality calculation results. The chart presents the top five average **betweenness centrality** values, categorized accordingly. On average, the locations with the highest centrality are those where people commonly transit to reach other destinations.

The presence of locations with high centrality value enables the existence of short paths between the nodes in the network. The obtained **diameter** was 6, a characteristic typical of small-world networks. Although the average path length is short, not all nodes in the network are reachable. The network is

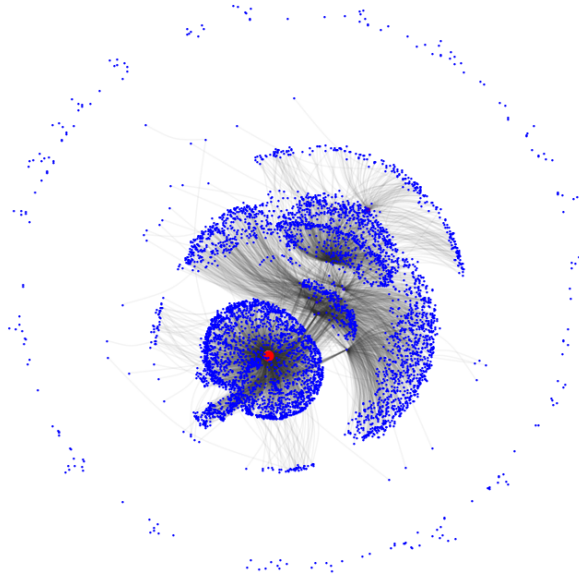


Fig. 4. Minimum spanning tree representation of the network.

fragmented into 180 **connected components** and contains 152 isolated nodes. It is possible to compare the global clustering coefficient (C) with the network density (ρ). A network is considered highly clustered when $C \gg \rho$. In the co-visitation graph, it was observed that $0.533 > 0.145$, where the giant component accounts for 96% of the network's nodes.

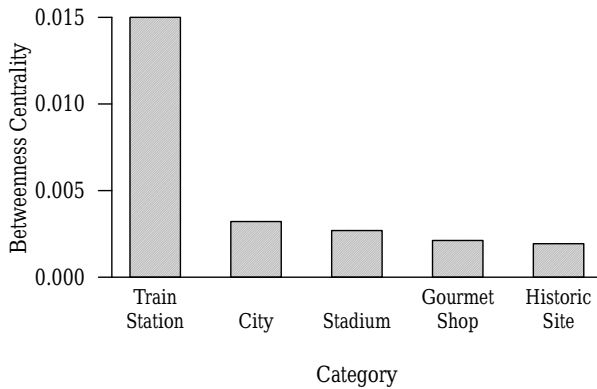


Fig. 5. Top 5 categories with highest average betweenness centrality.

The degree provides an estimate of the level of social activity between two locations. The average degree found in the network was 827.62. It is worth noting that about 80% of the nodes in the network had a degree lower than 500. This discrepancy occurs due to the presence of a small number of highly connected nodes (hubs), such as the *Colosseum*, which recorded the highest degree in the network (131,269). As shown in Fig. 6, the most connected nodes tend to have a higher probability of receiving new links. This is a typical feature of **scale-free networks**, where the degree distribution follows a power-law distribution. For the analyzed network, this behavior suggests that certain points of interest tend to

attract more visits, as a consequence of their popularity.

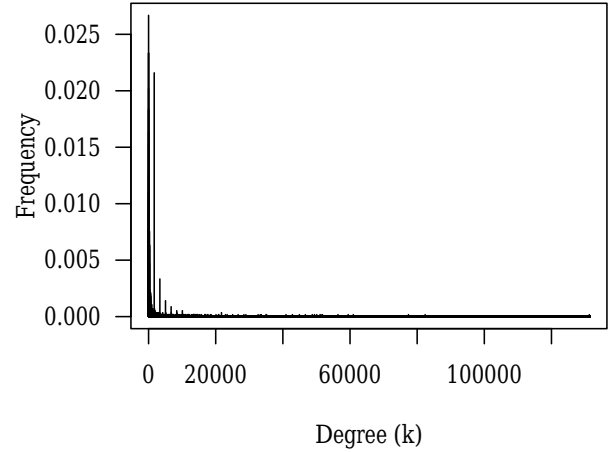


Fig. 6. Degree distribution of PoIs.

Table II shows network metrics after successive removal of top-centrality tourist nodes. Results highlight the key role these nodes play in preserving network connectivity and efficiency. The largest connected component (GCC) decreases by about 14%, indicating small fragmentation but overall network resilience. The diameter increases from 6 to 8, reflecting longer maximum paths and reduced efficiency. Density drops sharply, showing the network becomes sparser with fewer connections. The average shortest path length also rises, meaning nodes are slightly more distant. Tourist nodes concentrate a large number of connections, acting as important local hubs. However, even after their removal, the network remains resilient, maintaining overall connectivity and efficient communication paths among the remaining nodes. Notably, transport-related nodes (such as train stations) exhibit the highest centrality measures in the network and serve as crucial hubs that bridge distinct regions. Their presence helps maintain connectivity even as many tourist nodes are removed, contributing to the network's resilience.

TABLE II
NETWORK METRICS AFTER REMOVAL OF TOP CENTRALITY TOURIST NODES.

Removals	GCC Size	Diameter	Density	Avg. Shortest Path
None	5485	6	0.1452	2.2987
10%	5330	8	0.0881	2.4359
25%	5210	8	0.0693	2.4572
50%	5035	7	0.0629	2.4723
100%	4728	7	0.0631	2.4591

Table III shows network metrics after successive random removal of nodes, removing an equivalent number of nodes as in the targeted removal. Unlike the targeted removal of high-centrality tourist nodes, random removals result in only minor changes to the network structure. The diameter remains stable around 6, and network density fluctuates minimally near its original value. **These findings highlight the robustness**

of the network to random failures, contrasting with the significant structural changes caused by removing key tourist nodes.

TABLE III
NETWORK METRICS AFTER REMOVAL OF RANDOM NODES.

Removals	GCC Size	Diameter	Density	Avg. Shortest Path
None	5485	6	0.1452	2.2987
65	5421	6	0.1465	2.2998
163	5324	6	0.1449	2.3005
325	5165	7	0.1439	2.3194
650	4830	6	0.1476	2.3122

Next, we examine the behavior of users segmented according to their duration of stay in the city. This metric was calculated as the difference between the first and last check-in timestamps for each user. Records corresponding to users with only a single check-in were excluded from the analysis. The results presented in Fig. 7 demonstrate that user activity exhibits notable heterogeneity. **Approximately 57% of the users recorded check-ins over a period of 3 days.** This type of activity may reflect typical tourist behavior. According to data published by the Italian National Institute of Statistics (ISTAT), the average length of stay of tourists in accommodation facilities is 3.01 nights (2.66 nights for domestic tourists and 3.37 nights for non-resident tourists) [15].

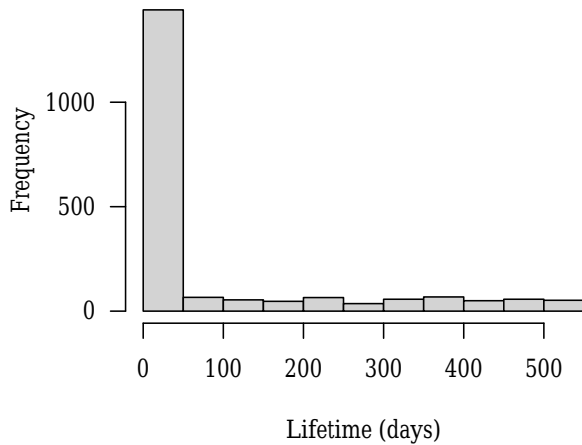


Fig. 7. Distribution of users' lifetimes.

From these results, we derived two networks from the original graph:

- 1) **Longest lifetime:** The graph was generated considering only the top 10% of users ranked by their longest lifetime in the dataset. On average, each user remained active for 454 days, with the most active user performing 1,684 check-ins. The resulting network is a connected graph with a high clustering coefficient, in contrast to the network formed by users with shorter lifetimes, which is fragmented.
- 2) **Shortest lifetime:** The graph was constructed using data from users with a duration of stay in the city not

exceeding three days. On average, each user stayed for 1 day and 30 minutes. This result does not necessarily imply that an individual physically stayed in the city for this duration. Some check-ins were recorded within seconds of each other. This behavior suggests that the user postponed sharing the check-ins to a time after the actual visit.

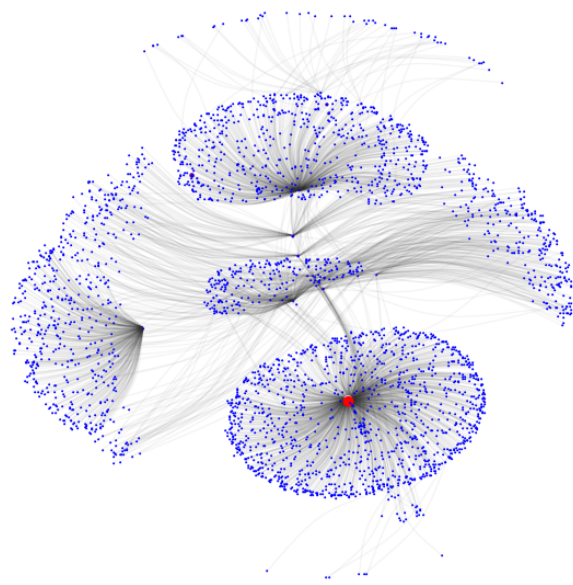
The structure of the network built with users of shorter duration differs drastically from the network with users of longer duration (see Fig. 8). In the longest lifetime network, there is a clear dominance of a single highly central node, the *Stazione Termini* (0.283), which serves as a major point of connectivity and likely concentrates a substantial share of movements between locations. In contrast, the second network exhibits a more distributed hub structure, with multiple prominent nodes, such as the *Colosseum* (0.177), the *Fontana di Trevi* (0.157), the *Stazione Termini* (0.154), and other landmarks. This indicates a more balanced distribution of centrality, where connectivity is supported by several key locations rather than being concentrated in a single point. Such a pattern may reflect the diversity of attractions visited and a more dispersed movement across the network.

Table IV summarizes the main differences between the two types of networks. The first network exhibits a higher level of connectivity compared to the second network. It suggests strong communities and repeated visits to similar groups of locations, reflecting patterns of routine and frequent socialization. In contrast, the second network exhibits lower density, fewer connections per location, and a larger diameter. These characteristics indicate more dispersed and less cohesive movement patterns, consistent with short-term visitors whose trajectories encompass multiple attractions rather than forming tightly interconnected location clusters.

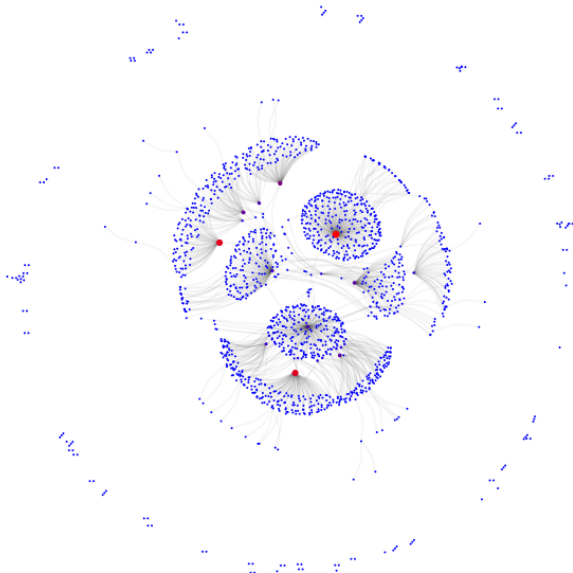
TABLE IV
CHARACTERISTICS OF THE DERIVED NETWORKS.

Metric	Longest lifetime	Shortest lifetime
Total check-ins	11,957	1,566
Total vertices	3,001	1,594
Total edges	1,968,667	15,197
Average degree	1,312.007	19.067
Average centrality	0.0004	0.0008
Clustering coefficient	0.71	0.17
Density	0.43	0.01
Diameter	4	7

In addition to the structural changes, it was also possible to observe a change in the semantics of the check-ins. Fig. 9 shows the five main categories that received the most check-ins. The largest lifetime network still contains check-ins at tourist locations. However, there is also a substantial number of check-ins at train stations, office sites, and coffee shops. In contrast, the second network shows a predisposition toward places frequented by tourists, such as historic sites, hotels, and monuments. This indicates that tourist attractions impact



(a) Longest lifetime network.



(b) Shortest lifetime network.

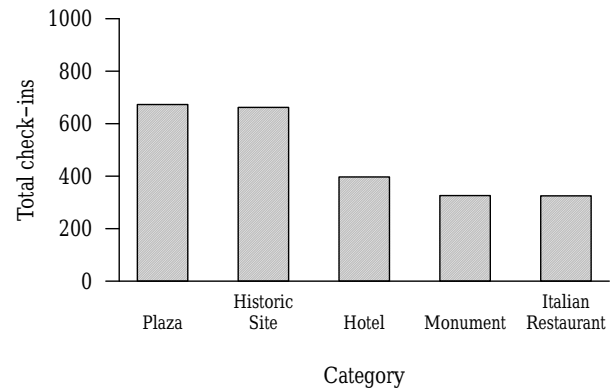
Fig. 8. Minimum spanning tree representation of the derived networks.

the co-visitation network differently depending on the visitors' length of stay.

Finally, we analyzed the degree distribution in the shortest lifetime network (Fig. 10). The results showed the presence of a small number of nodes with a high degree, in contrast to a larger number of vertices with a low degree. The maximum likelihood estimate of the exponent α is 2.174, consistent with values commonly found in real-world networks. The Kolmogorov-Smirnov (KS) test [16] was applied to verify the goodness of fit. At a 95% confidence level, the p-value obtained was 0.242, suggesting that the data do not signifi-



(a) Largest lifetime network.



(b) Shortest lifetime network.

Fig. 9. Total check-ins (top five categories) in the derived networks.

cantly deviate from a power-law distribution. This pattern is characteristic of preferential attachment, a mechanism where nodes with higher degrees tend to attract more connections over time.

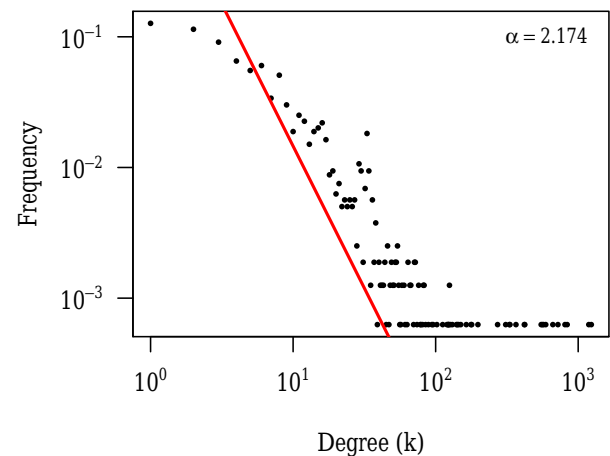


Fig. 10. Degree distribution in the shortest lifetime network (log-log scale).

Our analysis of the co-visitation networks suggests the presence of scale-free properties in the network, where popular

tourist places serve as central connectors linking various other locations, potentially including lesser-known sites. These hubs influence patterns of tourist flow and the sequence of visits. Disruptions at these key points (e.g., closures, overcrowding) could significantly impact overall mobility and visitor behavior. The results also revealed distinct visitation preferences across the analyzed networks. While tourist attractions attract substantial flows, Rome's main station persists as a critical transport hub, maintaining connectivity across diverse areas of the network.

These findings address the research question, supporting the alternative hypothesis that tourist attractions have a significant greater structural influence in the co-visitation network than non-tourist locations, exhibiting higher centrality (specially in the shortest lifetime network) and contributing more strongly to network connectivity. Although transport hubs play a vital role in preventing network fragmentation by acting as key bridges between regions, tourist nodes are essential for attracting new connections and promoting local clustering. Together, these elements create a balanced structure where transport hubs ensure global connectivity and resilience, while tourist locations drive the formation of dense, highly connected communities within the network.

VII. CONCLUSION

This study proposed an analysis of the structural properties of the check-in network of a tourist city. The experiments utilized a real-world dataset comprising check-in records collected over a period of 18 months. The objective of this study was to evaluate how the presence of tourist attractions influences the social dynamics within the city. Our analysis revealed that certain locations tend to attract a higher number of visitors, a pattern that persists even when considering scenarios dominated by local residents. This suggests that the attractiveness of these points of interest goes beyond tourism and significantly influences the daily social dynamics of the urban population.

Furthermore, the check-in network exhibits characteristic features commonly observed in complex networks, such as a low average shortest path length, which facilitates efficient node-to-node connectivity. In addition, the network exhibits scale-free behavior, evidenced by a heavy-tailed degree distribution and the emergence of hubs.

As future work, other scenarios could be evaluated and compared, such as cities with different social activity profiles. One potential research direction is the integration of the analyzed metrics with machine learning algorithms. This approach would enable the investigation of pattern discovery techniques and latent factors associated with the presence of tourist attractions.

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